

5 Marks Questions

Basics of Transport Layer

1. Define the Transport Layer.
2. List the functions of the Transport Layer.
3. What is process-to-process delivery?
4. What is end-to-end communication?
5. What is port addressing?
6. What is a port number?
7. Differentiate between port address and IP address.
8. What is segmentation?
9. What is reassembly?
10. What is multiplexing?
11. What is demultiplexing?

UDP

12. What is UDP?
13. List the features of UDP.
14. Write the UDP header format.
15. Mention the advantages of UDP.
16. Mention the disadvantages of UDP.
17. Give applications of UDP.

TCP

18. What is TCP?
19. List the features of TCP.
20. What is a TCP segment?
21. Write the TCP header format.
22. What is a sequence number?
23. What is an acknowledgment number?
24. What is flow control?

25. What is error control?
26. What is congestion control?
27. What is a sliding window?
28. What is checksum in TCP?

Connection Management

29. What is connection-oriented service?
 30. What is connectionless service?
 31. What is the three-way handshake?
 32. What is connection termination?
 33. What is a socket?
 34. What is a well-known port number?
 35. What is a dynamic port number?
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10 Marks Questions

Transport Layer Concepts

1. Explain the functions of the Transport Layer.
2. Explain process-to-process delivery.
3. Explain port addressing with examples.
4. Explain multiplexing and demultiplexing.
5. Explain segmentation and reassembly.
6. Explain end-to-end communication.

UDP

7. Explain UDP and its features.
8. Explain the UDP header format.
9. Explain the working of UDP.
10. Discuss the advantages and disadvantages of UDP.
11. Compare TCP and UDP.
12. Explain applications of UDP.

TCP

13. Explain TCP and its features.
14. Explain the TCP header format.
15. Explain sequence and acknowledgment numbers.
16. Explain flow control in TCP.
17. Explain error control in TCP.
18. Explain congestion control in TCP.
19. Explain the sliding window protocol.
20. Explain TCP segment structure.

Connection Management

21. Explain TCP connection establishment.
22. Explain TCP three-way handshake.
23. Explain TCP connection termination.
24. Explain socket programming concepts.
25. Explain port numbers and their types.

15 Marks Questions

1. Explain the Transport Layer in detail with its functions and diagram.
2. Explain process-to-process delivery and port addressing.
3. Explain multiplexing and demultiplexing with examples.
4. Describe the UDP protocol and UDP header format with a neat diagram.
5. Explain the working of UDP with advantages, disadvantages, and applications.
6. Explain the TCP protocol in detail.
7. Describe the TCP header format with a neat diagram.
8. Explain TCP connection establishment using the three-way handshake.
9. Explain TCP connection termination with a neat diagram.
10. Explain TCP flow control using the sliding window mechanism.
11. Explain TCP error control techniques.

12. Explain TCP congestion control mechanisms.
 13. Compare TCP and UDP in detail.
 14. Explain segmentation and reassembly in the Transport Layer.
 15. Explain port addressing and socket addressing in detail.
 16. Explain reliable data transfer in TCP.
 17. Explain sequence numbers and acknowledgment numbers in TCP.
 18. Discuss Transport Layer services and their importance.
 19. Explain the complete operation of TCP from connection establishment to termination.
 20. Explain congestion control algorithms in TCP.
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Frequently Asked Long Questions (15 Marks)

1. Transport Layer and its functions.
 2. TCP protocol and TCP header format.
 3. UDP protocol and UDP header format.
 4. TCP vs UDP comparison.
 5. Three-Way Handshake.
 6. TCP Connection Termination.
 7. Flow Control using Sliding Window.
 8. Error Control in TCP.
 9. Congestion Control in TCP.
 10. Port Addressing and Process-to-Process Communication.
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Frequently Asked Medium Questions (10 Marks)

1. Explain UDP.
2. Explain TCP.
3. Explain TCP Header.
4. Explain UDP Header.

5. Explain Port Addressing.
 6. Explain Flow Control.
 7. Explain Error Control.
 8. Explain Multiplexing and Demultiplexing.
 9. Explain Segmentation and Reassembly.
 10. Compare TCP and UDP.
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Frequently Asked Short Questions (5 Marks)

1. Define Transport Layer.
2. Functions of Transport Layer.
3. What is TCP?
4. What is UDP?
5. What is a Port Number?
6. What is a Socket?
7. What is Segmentation?
8. What is Reassembly?
9. What is Flow Control?
10. What is Error Control?
11. What is Congestion Control?
12. What is Sliding Window?
13. What is Three-Way Handshake?
14. What is Multiplexing?
15. What is Demultiplexing?
16. What is Process-to-Process Delivery?
17. What is Connection-Oriented Service?
18. What is Connectionless Service?
19. What is Sequence Number?
20. What is Acknowledgment Number?

Exam Priority (Most Important Topics)

Very Important (15 Marks)

- Transport Layer Functions
- TCP Protocol
- TCP Header Format
- UDP Protocol
- UDP Header Format
- TCP Three-Way Handshake
- TCP Connection Termination
- TCP Flow Control (Sliding Window)
- TCP Congestion Control
- TCP vs UDP

Important (10 Marks)

- Port Addressing
- Multiplexing & Demultiplexing
- Segmentation & Reassembly
- Error Control
- Reliable Data Transfer
- Socket Addressing

Short Notes (5 Marks)

- Port Number
- Socket
- Flow Control
- Congestion Control
- Sequence Number
- ACK Number
- Three-Way Handshake

- UDP Features
- TCP Features
- Sliding Window Protocol